

# Colton Kawamura

Software Engineer & Ph.D. Candidate

Franklin, NC  
Top Secret / SCI Clearance

## Education

- Present **Ph.D. in Physics**, *University of Amsterdam*, Amsterdam, NL  
Theoretical and Computational Analysis of Disordered Condensed Matter Systems.
- 2020 **M.S. Physics**, *Naval Postgraduate School*, Monterey, CA  
GPA 3.98. Selected coursework: Advanced Graduate Mathematics, Quantum Mechanics, Machine Learning, Linear Systems.
- 2012 **B.S. Applied Physics, Pre-Med**, *U.S. Naval Academy*, Annapolis, MD  
GPA 3.52. Research in computational physics.

## Skills

- Languages C#, Python (SciPy, NumPy, Pandas), MATLAB (Simulink), C++, Julia, R, LaTeX
- Tech .NET, HPC, Linux, Bash, Parallel Computing, Git, SVN, Jenkins, SLURM, Gaussian, LAMMPS, GPAW

## Experience

- 2025 – Present **Software Engineer**, *Duotech Services*, Franklin, NC  
Program engineer for software-defined defense aerospace automated test equipment (ATE).
  - Principal software engineer for design and development of ATE software from conception through launch and delivery, including integration into hardware systems.
  - Lead for DevOps pipelines and reliability (CI/CD, monitoring, observability).
  - Integrated AI into software systems, including local on-prem deployments to accelerate software development.
  - Integrated software with hardware for automated test and evaluation systems in defense deployments.
- 2024 – 2025 **Research Faculty, Computational Mathematics & Physics**, *Naval Postgraduate School, Physics Dept.*, Monterey, CA / Remote
  - Built high-performance C++ molecular dynamics simulation frameworks.
  - Designed parallel software architectures, cutting runtime by orders of magnitude.
  - Implemented FFT-based tools for trajectory and signal analysis (filtering, phase extraction).
  - Published work on a Generative AI approach to counter-swarming using neural networks.
- 2020 – 2024 **Technical Program Manager**, *US Navy, US Africa Command*, Stuttgart, Germany  
Explosive Device & Unmanned Aerial System Defense.
  - Analyzed large unstructured data to extract trends and predict counter-drone missions.
  - Managed UAV and electronic countermeasure programs across multiple countries under Agile.
  - Coordinated deployment of counter-UAS systems with host nations in combat environments.
  - Direct advisor on portfolio to Commander, US Africa Command.
- 2018 – 2020 **Graduate Student, Quantum Thermochemistry**, *U.S. Naval Postgraduate School*, Monterey, CA
  - Researched explosive thermodynamic properties using computational quantum thermochemistry.
  - Completed interdisciplinary Undersea Warfare curriculum (sensors, weapons, mathematics, physics, engineering, operations analysis).

- 2012 – 2018 **Special Operations (EOD) Officer**, *US Navy, EOD Mobile Unit ONE & THREE; SEAL Team THREE*, Various Locations
- Conducted 23+ combat missions and over 200 underwater reconnaissance dives across multiple operational areas.

## ██████████ Awards & Publications

- Academic Chief of Naval Operations Undersea Warfare Award; Outstanding Thesis Award
- Military Bronze Star Medal; Combat Action Ribbon; Joint Service Achievement Medal; and others
- Recent Pub. *High-Energy Density Material with Magnetically Modulated Ignition*, JASC (2024)